DHRUV SAINI

dhruvs.pages.dev @ dhruv9saini@gmail.com

EXPERIENCE

Arx Labs
- Built PCB design automation workflows for placement and routing
- Validated native KiCad ERC/DRC/CAM outputs for PCB design artifacts and training data

A Sustainable Future

CEO | Oct 2023 - Present
- Built and scaled the **PCM**, an AI-powered mobile app predicting paper use, to **30,000 students** nationwide in over 150 schools
- Trained a symbolic regression model with **PySR** to predict paper consumption based on school demographics
- Leading **a team of 20** to develop a nonprofit website and iterating on AI model, now helping save 1.7M sheets of paper per year
- Received **$30k** in grants supporting sustainability impact

Dimension

Software Engineer Intern | Jun 2025 - Sept 2025
- Implemented webhooks to allow external companies to integrate with the enterprise collaboration platform
- Led the process to allow Stripe, Vercel, Github, and many other hosted bots in channels
- Revamped and modernized the UI, completing several refactors that culminated in a **70%** reduction of load time

University of Washington

Quantum Technical Writing Intern | Aug 2024 – Sept 2024
- Worked with Prof. Anantram to develop an editing workflow for his book *Quantum Mechanics for Scientists and Engineers*
- Used Python and React to automatically compare textbook versions and allow students to easily select the best version of any paragraph
- Accelerated editing time by **15x**

PROJECTS

CAD-bench: Benchmarking Language Models on Functional CAD Generation
- Built a benchmark for CAD generation that evaluates whether parts satisfy functional requirements, not just renderability or visual similarity
- Executed rigid-body simulations for complex mechanisms such as gearboxes to verify that generated CAD parts function as intended
- Accepted to the ICML AIWILD and FAGEN workshops; currently under review at NeurIPS, with code available here and paper available here

EDA Bench: An Execution-Based Benchmark for Language Model Agents on Functional PCB Construction
- Built a benchmark for PCB generation that evaluates electrical behavior, not just whether generated boards build successfully
- Integrated KiCad, routed-connectivity extraction, ERC/DRC checks, ngspice simulation, and external I/O specifications to verify circuit functionality
- Accepted to the ICML FAGEN workshop; currently under review at NeurIPS, with code available here and paper available here

ThinkSwitch: Iterative LoRA training and Weight Interpolation for Low-Compute Improvement on Specific-Purpose Reasoning Tasks
- Applied weight interpolation and LoRA to iterative distillation and amplification to greatly improve Qwen3-4B AIME score with only $3 in compute
- Awarded 1st place at the Washington State Science and Engineering Fair, preprint available here

Muon Browser | *Typescript*, *Electron*
- Created a web browser that allows the user to open and browse pages and notes on an infinite canvas
- Available to download and use here

Residual Controllers: Steering Thinking Modes in Open Language Models
- Built fixed residual-stream controllers that steer open language models between thinking and direct-answer response modes
- On AIME 2026, a one-step controller entered thinking mode on all 30 problems and scored 16/30, versus 1/30 with thinking off and 2/30 with random control; paper here

AWARDS AND HONORS

National Science Bowl

Team Captain | 2021-Present
- Founded Bellevue High School's first science bowl team, winning 1st place at the nationwide virtual competition
- Won a fully-funded trip to Washington D.C. for the National Finals

USAAIO

Competition | 2024-Present
- National Finalist, Bronze Medal
- Competition covered AI theory and programming in machine learning, deep learning, transformers, natural language processing, computer vision, and generative AI

ARC - The American Rocketry Challenge

Team Captain | 2025
- Captained a team of five to build a model rocket following specific criteria, qualified for National Finals twice

EDUCATION

Bellevue High School
- 36 ACT, 100+ hours of community service, DECA ICDC finalist
- President of Programming Club, coach at Tyee MS and Odle MS programming clubs